

Machine Learning in Python

Introduction to Machine Learning

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- 1 Introduction to Machine Learning
- 2 The Machine Learning Workflow
- 3 Train-Test Splits and Cross-Validation
- 4 Exploratory Data Analysis (EDA)
- 5 Data Preprocessing
- 6 Feature Engineering

What is Machine Learning?

Machine Learning

Definition

ML vs Traditional Programming:

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- ML: input + output $\rightarrow$ rules (the model)

The Machine Learning Workflow

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- Evaluation
- Deployment and Monitoring

Train and Test Datasets

Splitting the dataset

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Why Split?

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Splitting the dataset

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- To prevent overfitting, where the model learns noise in the training data instead of the underlying pattern.
- The **test** dataset is put aside and not used until the end, when the model is evaluated.

Cross-Validation

The **train** dataset can be further split into **training** and **validation** sets to tune the model's hyperparameters and prevent overfitting.

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- **Stratified K-Fold:** Ensures that each fold has the same proportion of classes as the entire dataset, which is particularly useful for imbalanced datasets.

Data Leakage

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The Exam Analogy

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The Exam Analogy

A student who accidentally sees the answer key scores 100%, but learns nothing and fails the next exam.

Common Sources of Leakage

Common Sources	Example
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- **Duplicate samples**: Same subject in both train and test (multiple measurements per patient).

Preventing Leakage

Prevention Strategies

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- **Think causally:** "Would I have this feature at prediction time?"

Exploratory Data Analysis (EDA)

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Exploratory Data Analysis (EDA) is the process of analyzing datasets to summarize their main characteristics, often using visual methods.

Exploratory Data Analysis (EDA)

Purpose of EDA

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EDA Techniques

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- **Missing Value Analysis:** Identifying and handling missing data.

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- **Correlation Analysis:** Identifying relationships between variables.
- **Missing Value Analysis:** Identifying and handling missing data.
- **Outlier Detection:** Identifying and handling outliers in the data.

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Data Preprocessing is the process of transforming raw data into a format suitable for analysis and modeling.

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- **Encoding Categorical Variables:** Techniques include one-hot encoding, label encoding, or using embeddings for high-cardinality categorical variables.

Feature Engineering

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Feature Engineering is the process of using domain knowledge to select, modify, or create features (variables) that improve the performance of machine learning models.

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- **Feature Transformation:** Techniques include logarithmic transformation, polynomial features, and interaction terms.
- **Feature Creation:** Creating new features based on existing ones, such as aggregating time series data, extracting date/time components, or creating domain-specific features.